

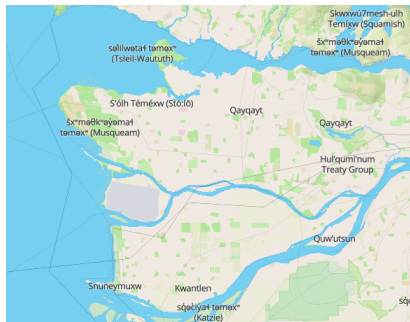
ELEC 221 Lecture 01

Overview and intro to signals and systems

Thursday 10 September 2026

Land acknowledgement

The land on which we gather today is the traditional, ancestral, and unceded territory of the Musqueam People.



Explore more:

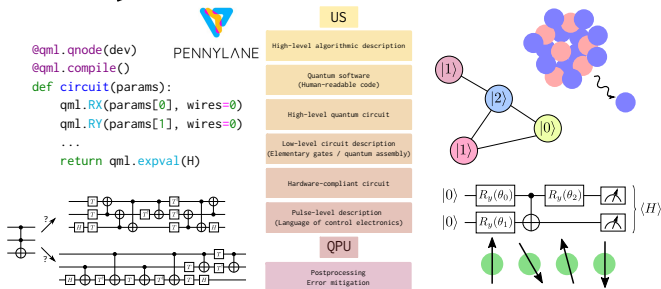
- <https://www.musqueam.bc.ca/>
- <https://native-land.ca/>

Intros

Olivia Di Matteo — olivia@ece.ubc.ca

I have a physics background. My group works on open-source quantum software and algorithms.

>QSAR QUANTUM SOFTWARE & ALGORITHMS RESEARCH



Signals and systems are EVERYWHERE: audio processing

- how do equalizers and effect pedals work?
- how do you synthesize digital sounds?
- how does active noise cancellation work?

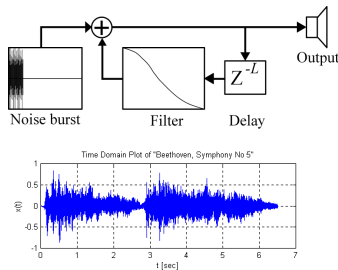


Image credits:

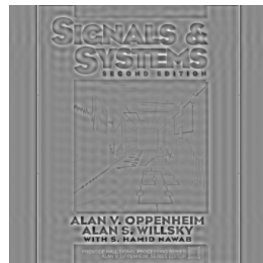
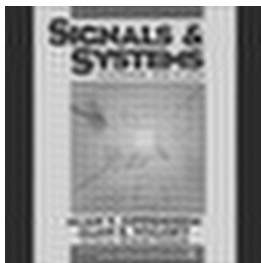
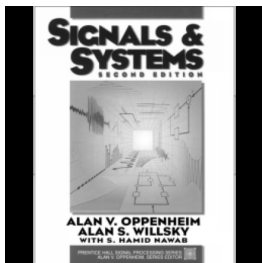
<https://www.uberchord.com/blog/guitar-effects-pedalboard-order/>

<https://en.wikipedia.org/wiki/File:Karplus-strong-schematic.svg>

<https://stats.stackexchange.com/questions/72386/find-music-tempo-with-a-audio-signal>

Signals and systems are EVERYWHERE: image/video processing

- why do wheels in old movies spin backwards?
- how do photo filters work?
- how can we improve the quality of blurry or noisy images?



Signals and systems are EVERYWHERE: data processing and analysis

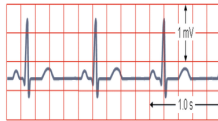
- how do we encode and transmit data correctly and efficiently?
- how do we identify trends and make predictions from data?
- how do we engineer systems that adapt based on input data?



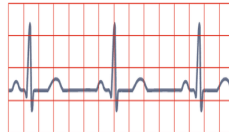
Image credit: google.com market summary

Signals and systems are EVERYWHERE: life sciences

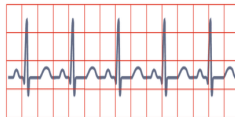
- how do we engineer better hearing aids?
- how does medical imaging work?
- how do we process and interpret the results of medical tests?



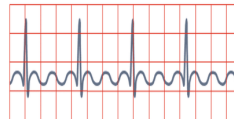
(a) sinus rhythm, normal complex, Heart rate-60-100/min



(b) bradycardia, normal complex, hear rate <60



(c) tachycardia, normal complex, heart rate >100/minute



(d) Atrial flutter, the irregular response of ventricular, rapid flutter waves

Image credit: Thirugnanam, Mythili & Pasupuleti, Megana. (2021). Cardiomyopathy-induced arrhythmia classification and pre-fall alert generation using Convolutional Neural Network and Long Short-Term Memory model. Evolutionary Intelligence. 14. 10.1007/s12065-020-00454-0.

Course learning outcomes

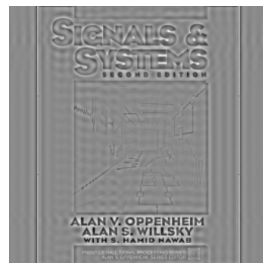
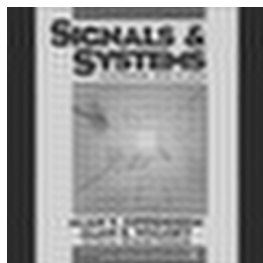
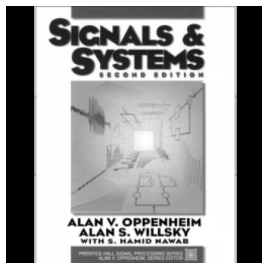
A broad overview of the processing and analysis of signals and systems, grounded in *real-world examples*.

- Express continuous- and discrete-time signals as functions, and characterize how linear, time-invariant systems respond to them
- Apply Fourier analysis (and generalized transforms) to determine key properties of signals and systems in the time and frequency domain
- Design systems that modify signals using filters, sampling, modulation, and feedback

By the end of the course you'll be able to do the above both **mathematically** and **computationally**.

Course overview

Recommended text: *Signals & Systems* 2e, Oppenheim & Willsky.



All lecture slides/demos will be posted on the course GitHub:

<https://github.com/glassnotes/ELEC-221>

If you have questions...

In class: questions always welcome!

My office hours (KAIS 3043):

- Fridays 12:00-12:50
- open-door after 12pm on weekdays
- by appointment

Tutorials: Monday 17:30-19:30. Facilitated TA team (contact details in syllabus).

Piazza: find link in Canvas.

Course overview: assignments and grading

- 10% In-class quizzes (10)
- 5% Tutorial assignments (5)
- 25% Midterm exam
- 60% Final exam

Course overview: assignments and grading

(10%) In-class quizzes (10)

- In PrairieLearn during first 10 mins. of Tuesday classes (calendar in syllabus)
- 1-3 auto-graded questions based on previous week
- Individual; open-book, no AI

Quiz 1 this Tuesday.

No deferred quizzes. Lowest quiz grade automatically dropped.

Course overview: assignments and grading

Half of tutorial sessions are problem-solving, others are hands-on signal processing with Python.

(5%) Tutorial assignments (5)¹

- Distributed and submitted on PrairieLearn
- Done in groups of up to 4; start in tutorial, due later

Calendar in syllabus. First tutorial this **Monday** (problem solving); first tutorial assignment on 21 September (Python basics).

No late submission. Graded for reasonable effort and completion.

¹Developed by ENPH students, funded by UBC Open Educational Resources Implementation Grant!

Course overview: assignments and grading

(0%) Technical assignments (4-5)

- Distributed on PrairieLearn
- Mix of numerical input and pen-and-paper
- **For practice only.** Solutions will be shared 1-2 weeks later.

See also end of lecture slides for suggested textbook questions.

Course overview: assignments and grading

(25%) Midterm: Monday 2 Nov. (during tutorial)

- Two-stage exam: solve alone first, then redo subset of question in small groups
- Weighted avg. of individual (85%) and group (15%) scores

(60%) Final exam (date TBD)

- Individual exam
- **Cumulative**, emphasis on later parts

Today

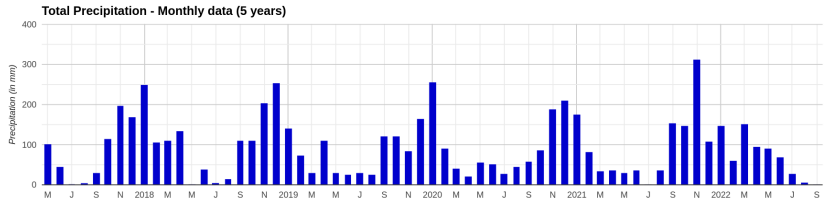
Learning outcomes:

- Define and identify real-world examples of signals and systems
- Express continuous-time and discrete-time signals as functions
- Apply time shift, scaling, and reversal transforms to signals
- Identify if a system is causal, memoryless, stable, or invertible

Signals

Signals are “patterns of variations” that contain information about the behaviour of a phenomenon.

Example: total precipitation in Vancouver vs. time



Mathematically: functions of one or more independent variables (often, but not always, time).

Image generated from: <https://vancouver.weatherstats.ca/charts/precipitation-monthly.html>

Signals

In this class we will be

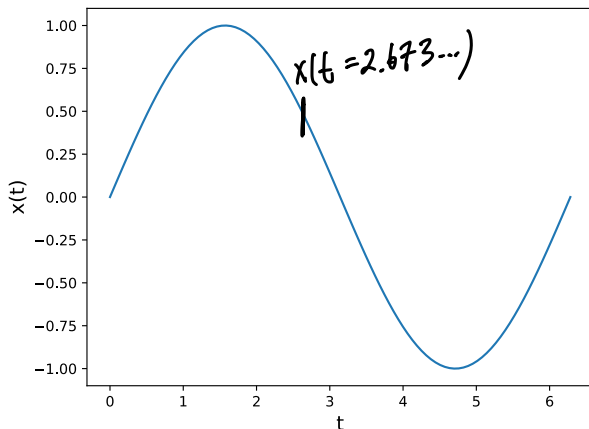
- Transforming signals
- Synthesizing and analyzing signals
- Determining how systems respond to them
- Designing systems to manipulate signals in specific ways

We consider two types of signals:

- ① Continuous-time (CT)
- ② Discrete-time (DT)

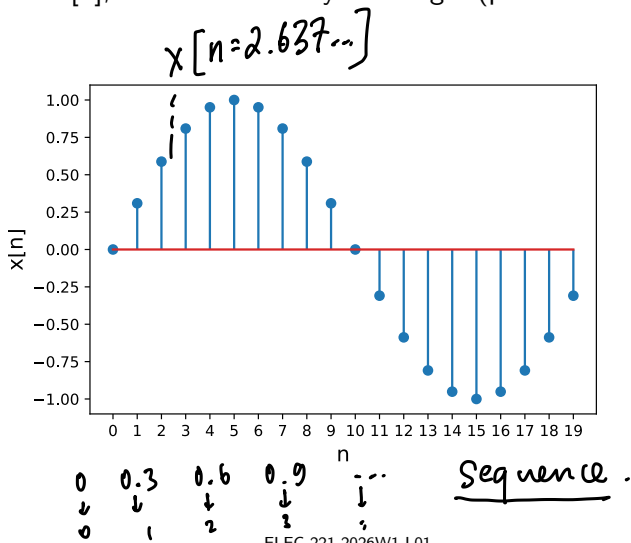
Continuous-time (CT) signals

Notation: $x(t)$, where t is any real number.



Discrete-time (DT) signals

Notation: $x[n]$, where n is strictly an integer (positive or negative).



Exercise: CT vs. DT signals

Are the following signals CT or DT?

Signal	Type
radio broadcast	CT (DT: internet radio)
seismogram	CT
<u>daily</u> temperature high	DT (1x/day)
speed of a car	CT
a movie	DT

Transformations of independent variables

$$x(t)$$

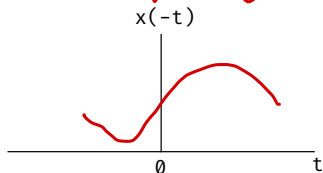
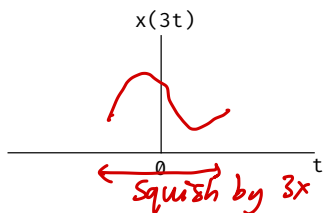
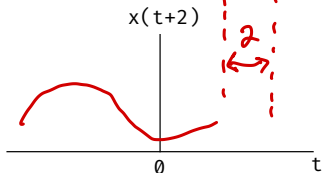
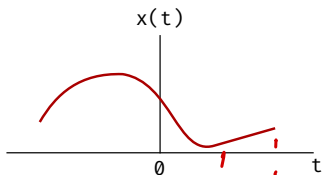
Most of what we do involves *transforming* signals.

Simplest example: transform the independent variable...

- Time shift: $x(t) \rightarrow x(t - t_0)$ shift right by t_0
- Time scaling: $x(t) \rightarrow x(at)$ a : scaling factor
- Time reversal: $x(t) \rightarrow x(-t)$ $|a| > 1$ faster
 $|a| < 1$ slower

Intuition: playing music slower/faster, or in reverse.

Exercise 1: Transformations of independent variables



Transformations of independent variables

These transforms can also be *composed*:

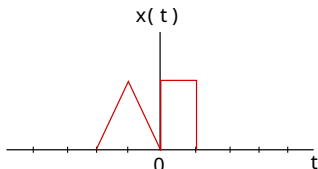
$$x(t) \longrightarrow x(\alpha t + \beta)$$

Transform in order: “*shift, scale, reverse*”

- ➊ Shift by β
- ➋ Scale by $|\alpha|$
- ➌ Reverse if $\alpha < 0$

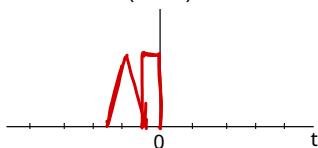
Exercise 2: Transformations of independent variables

$$x(t) \rightarrow x(-2t + 1) = x(-2(t - 1/2))$$

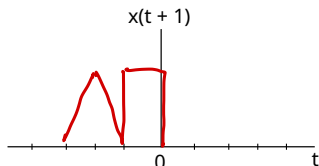


scale

$$x(2t + 1)$$

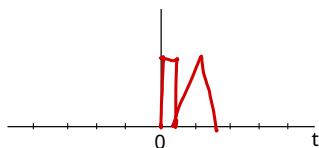


shift.



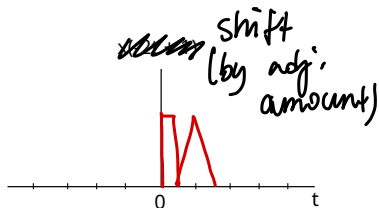
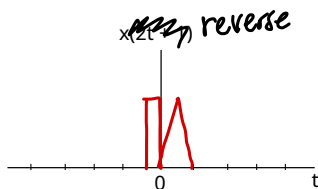
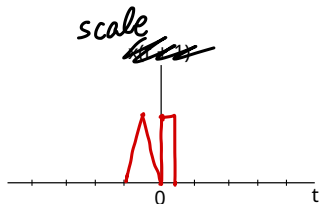
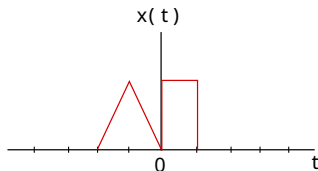
reverse.

$$x(-2t + 1)$$



Exercise 2: Transformations of independent variables

$$x(-2t+1) = x(-2(t - 1/2))$$



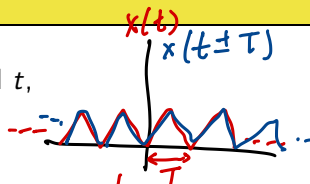
Consequences: periodicity and symmetry

A CT signal is periodic with period T if for all t ,

$$x(t + mT) = x(t)$$

↳ integer m

The smallest T for which this holds is the **fundamental period**.

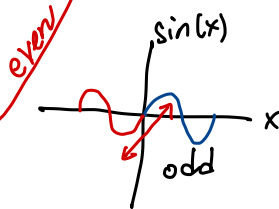


A CT signal is **odd** if for all t

$$x(-t) = -x(t)$$

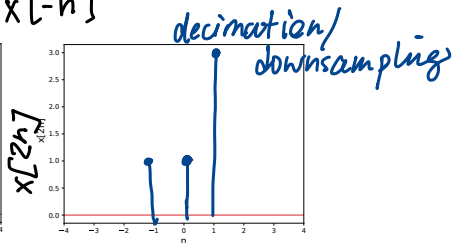
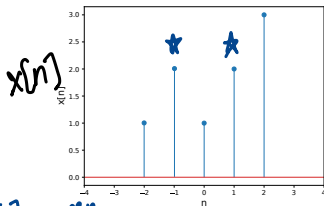
and **even** if

$$x(-t) = x(t)$$



Exercise: DT

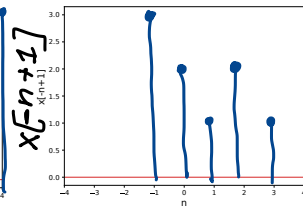
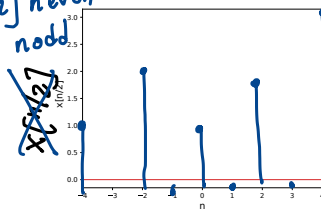
- Time shift: $x[n] \rightarrow x[n-n_0]$
- Time scaling: $x[n] \rightarrow x[an]$
- Time reversal: $x[n] \rightarrow x[-n]$



$x[n] \rightarrow \begin{cases} x[n/2] & \text{even} \\ 0 & \text{odd} \end{cases}$

zero-insertion

~~$x[n/2]$~~



Consequences: periodicity

A DT signal is periodic with period N if for all n ,

$$x[n+N] = x[n]$$

The smallest N for which this holds is the **fundamental period**.

A DT signal is **odd** if for all n

$$x[-n] = -x[n]$$

and **even** if

$$x[-n] = x[n]$$

Systems

Systems respond to signals and produce some desired behaviour, or other signals as outputs.

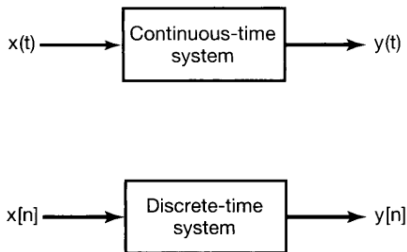
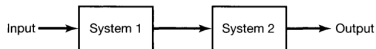


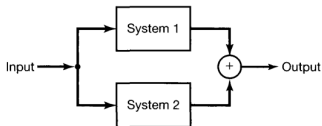
Image credits: Signals and Systems 2nd ed., Oppenheim

Systems: block diagrams

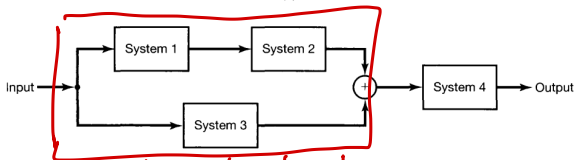
We can combine systems in series or parallel.



(a)



(b)



Big combined system

Image credits: Signals and Systems 2nd ed., Oppenheim

Properties of systems

- ① Memory
 - ② Invertibility
 - ③ Causality
 - ④ Stability
 - ⑤ Time invariance
 - ⑥ Linearity
- } Tuesday.

Properties of systems: memory

A system is **memoryless** if the output at each time depends only on the input *at the same time*.

(✓) Output voltage after current goes through a resistor (R)
(memory less)

$$V(t) = R I(t)$$

output signal current

(✗) Output voltage after current $I(t)$ goes through capacitor (C)

$$V(t) = \frac{1}{C} \int_{-\infty}^t I(\tau) d\tau$$

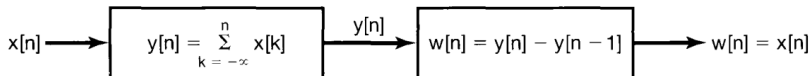
(✗) A delay system



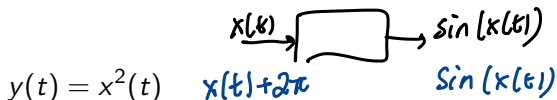
Properties of systems: invertibility

A system is **invertible** if distinct input signals lead to distinct output signals. An invertible system has an **inverse system**.

(✓) Accumulator



(✗) The system



Careful: must consider entire signals, not just signal values.

Image credits: Signals and Systems 2nd ed., Oppenheim

Properties of systems: causality

A system is **causal** if the output at *any time* depends only on the input at the present time or in the past.

(✓) Output voltage after current $I(t)$ goes through capacitor (C)

$$V(t) = \frac{1}{C} \int_{-\infty}^t I(\tau) d\tau$$

(✗) A moving average



$$\rightarrow y[n] = \frac{1}{2M+1} \sum_{k=-M}^M x[n-k]$$

(✗) Time reversal

$$y[n] = x[-n]$$

Properties of systems: stability

A system is **stable** if small changes in input do not cause the output to diverge (bounded inputs lead to bounded outputs).

(✓) Moving average of a bounded function, $x[n] \leq B$

$$y[n] = \frac{1}{2M+1} \sum_{k=-M}^M x[n-k]$$

(✗) Compound interest in a bank account

$$y[n] = 1.01 y[n-1] + x[n]$$

For next time

Next two classes on Zoom

Content:

- Quiz #1 (transformations of variables, system properties)
- Linear time-invariant (LTI) systems, DT unit impulse, impulse response, and convolution sum

Action items:

- 1 Follow instructions on Canvas for PrairieLearn and Piazza
- 2 Prepare for tutorial assignment 1 on Monday

Recommended reading:

- from today's class: Oppenheim chapters 1.1-1.2, 1.5-1.6
- practice problems: 1.4, 1.5, 1.15, 1.21a-d, 1.22a-d, 1.30a-c,e-g,k-n
- for next class: Oppenheim 1.4, 2.1-2.3